



Gollum was brought to life thanks to MoCap and Andy Serkis

According to Rajnish Sah, Manager Operations, "The rig is the only one of its kind in the country." It has added zing to several ad-films, Hindi feature films, television serials, documentaries and music videos. You can see the Motion Control effect in the Titan Raga watch advertisement with multiple Gul Panags, and also in the Palmolive Aroma talcum powder, and the LML Freedom bike commercials. "The French company that had commissioned the Palmolive advertisement were so impressed by the Indian version that they flew the model down to India, and the international version was also shot here," says Sah. (Find the actual clips on the Digit DVD.)

Prime Focus is also responsible for the *Mujse Shaadi Karoge* scene. Motion Control was used there to replicate the six Akshay Kumars. (See the movie clip on the Digit DVD.)

Prime Focus has a seven-member team, led by Manager Sandeep Sethi, to run the Motion Control rig. The equipment is powered by 'Flair'—a software that helps the operator build complex moves and coordinate all the axes of the Motion Control system. "Like MoCap software, Flair is compatible with most animation software such as Maya and 3D Studio Max," says Sethi.

The Motion Control equipment can track, rotate, lift, pan, tilt, roll, zoom, etc., when shooting a scene. Its geometry allows it to perform these moves from ground level to a maximum of 4.1 metres (13.5 feet) with minimal image jerkiness. The rig can repeat a complex series of such moves with the flick of a button—

something no human cameraman could ever achieve. It is also used to replicate crowds, make objects appear and disappear (the Hindi movie *Gayab* was filmed using Motion Control), and juggle background imagery.

Motion Control has been employed in the making of films such as *Harry Potter and the Prisoner of Azkaban*, *Alexander*, *The Matrix (Reloaded and Revolutions)*, *James Bond: Die Another Day*, *Stuart Little 2*, *Terminator 3*, and *Black Hawk Down*.

Techniques like Motion Capture and Motion Control represent the acme of synergy between technology and human creativity. MoCap makes it possible to create fluid, life-like character animation, in a faster, easier and more realistic manner than traditional animation. Motion Control helps capture near-impossible camera angles to add a dash of the unreal to cinematic action.

Whereas MoCap technology in its current avatar can record only the movement of a few key points on the actor's body (where the reflective markers are placed), in the not-so-distant future, MoCap sessions might be able to capture the complete frame of a person, from several angles. Every inch of the actor's body can be mapped, thus providing animators MoCap data with higher resolution of detail.

However, all the advanced equipment in the world will not create a *Matrix*, or a *LoTR*—that requires the ability to put such equipment to good use, and of course, some out-of-the-box thinking—Wachowski *ishtyle!* ☒

meera_vankipuram@thinkdigit.com

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